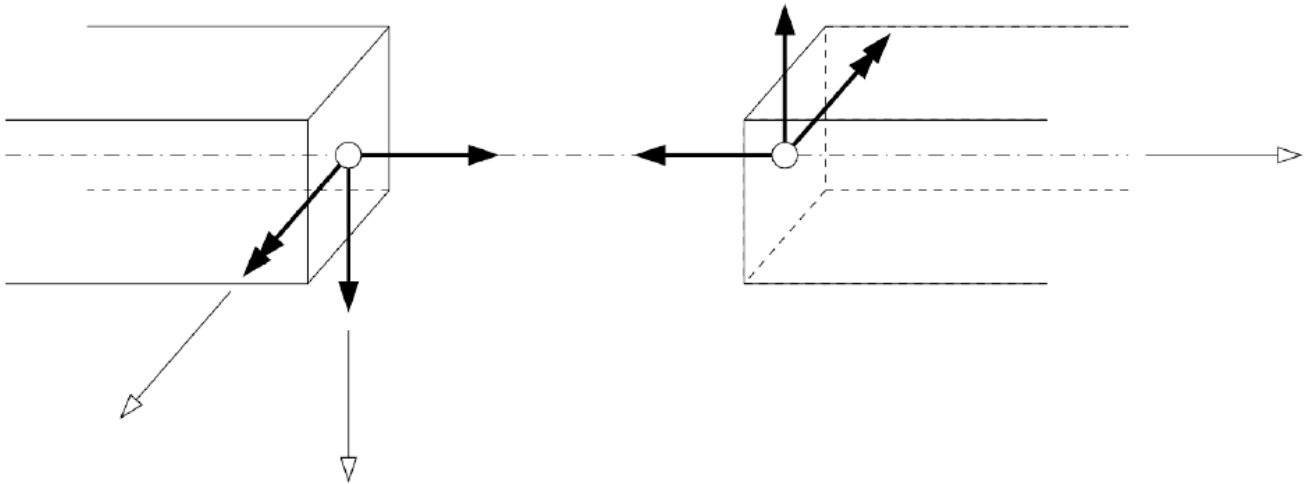
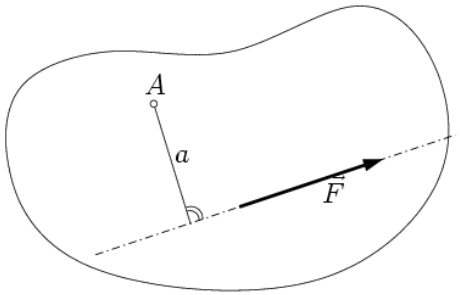
Engineering Mechanics I

Lecture



4.4.1. Moment of a Force



- Force $\vec{F}$ generated with respect to A:

$$M_A = aF\,[Nm] \text{ z.B. } \left(\circlearrowleft\circlearrowright +-\right).$$

- define positive direction of rotation
- distance *a* is named **lever arm**
- line volatility remains
- Moment of F bound to a reference point

Resulting moment of multiple forces:

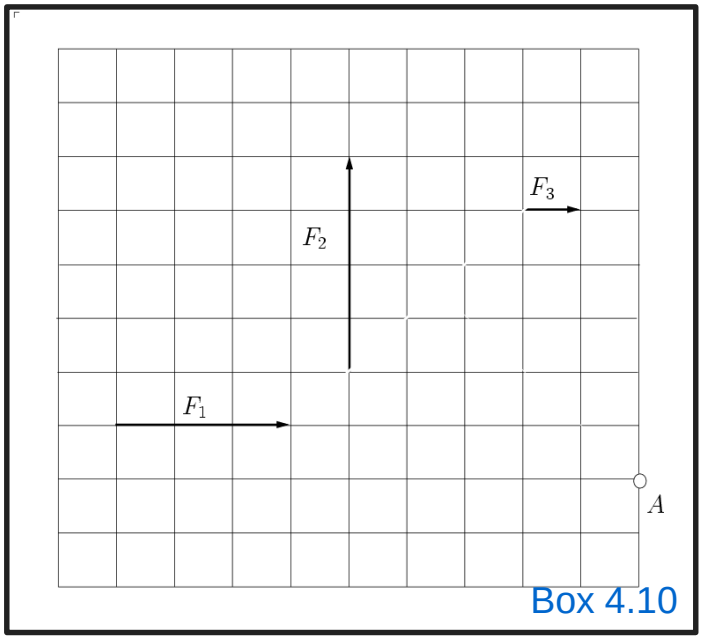
Box 4.8

4.4.1. Moment of a Force

Example 4.1: Numerical example for the moment

Required: Moment of forces F_1 to F_3 with respect to A.

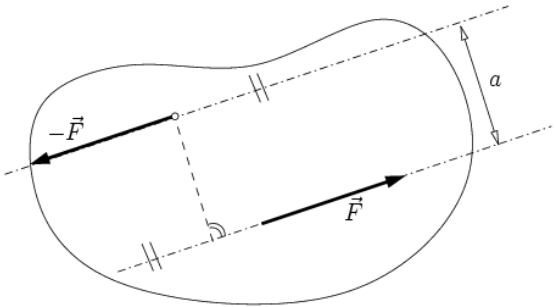
Box 4.9



Box 4.10

4.4.2. The force pair

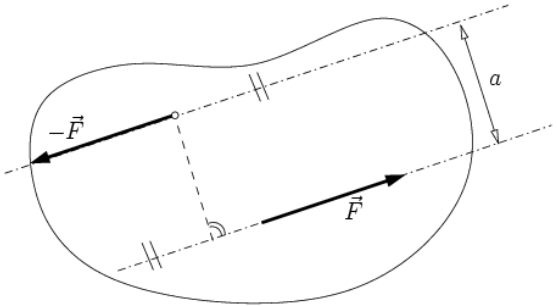
- Same magnitude,
- with **opposite** directions and
- are on **parallel** lines of action.



- $\vec{R}$ = 0, but rotational effect
- Construction of a resulting force vector not possible

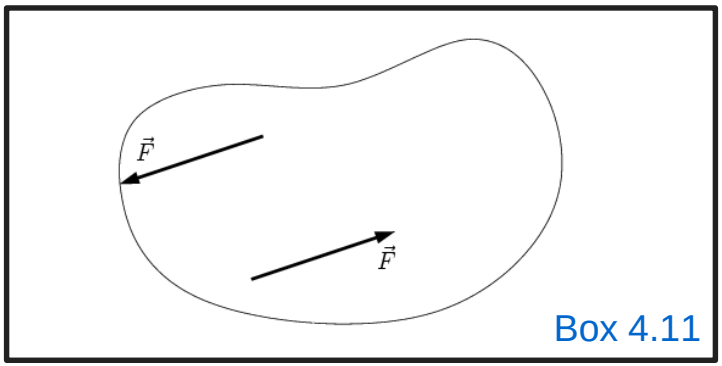
4.4.2. The force pair

- Description of the moment action of a pair of forces by:
- Magnitude: $M = aF \quad [Nm]$
- Direction or sense of rotation: + or -



4.4.2. The force pair

- A force pair is defined by specifying the **magnitude** M ("force times lever arm") and the **direction of rotation**. It cannot further be reduced.
- The moment of a pair of forces is equal to the sum of the moments of the two individual forces with respect to **any** reference point.



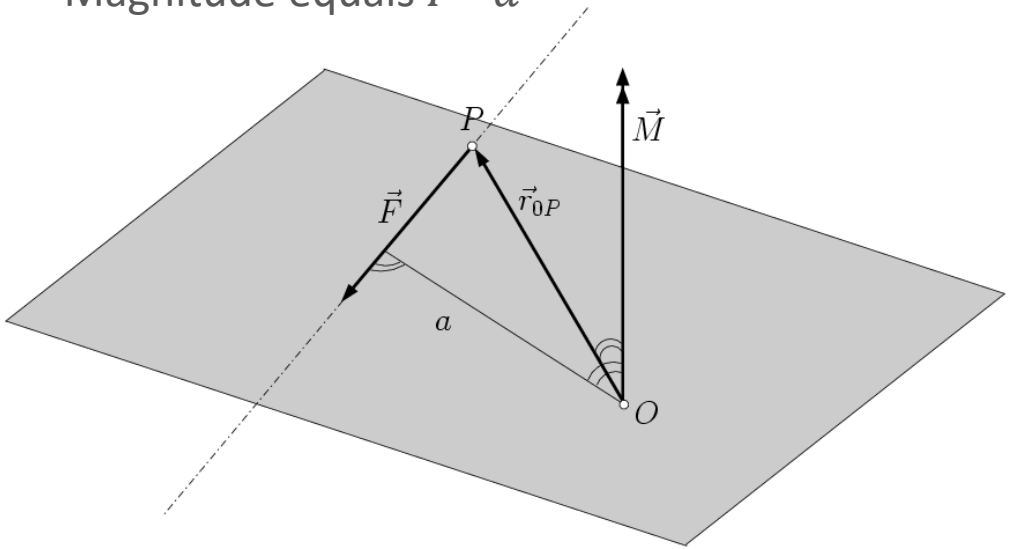
Box 4.11

For any A it holds:

Box 4.12

4.4.3. Moment as a vector

- Moment vector is perpendicular to the effective plane
- Direction of a right screw
- Magnitude equals $F \cdot a$

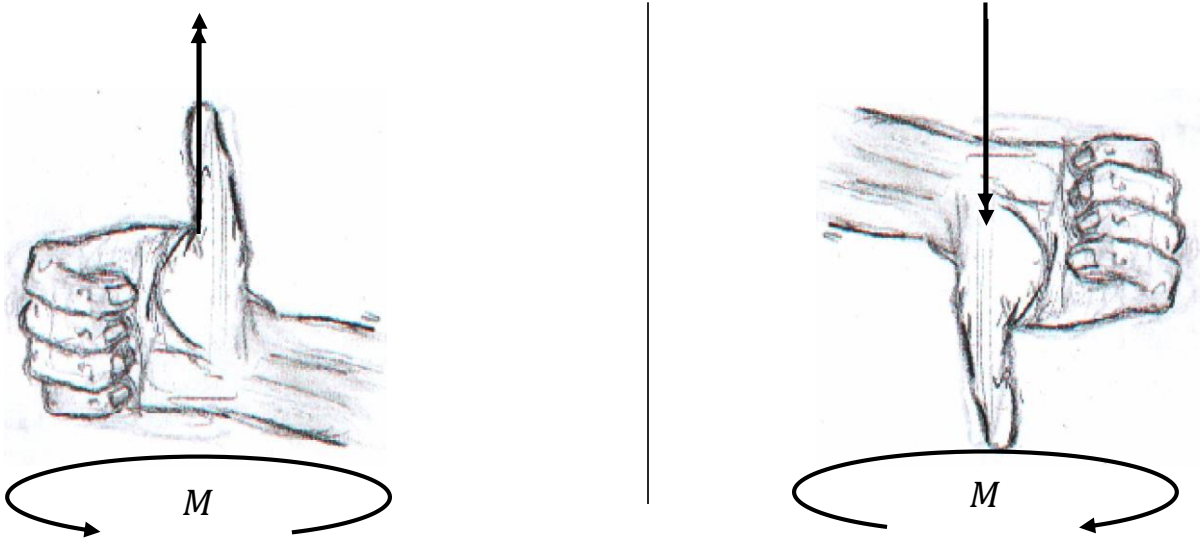


Calculation of the moment from the **cross product** of force and position vector of the point of application:

Box 4.13

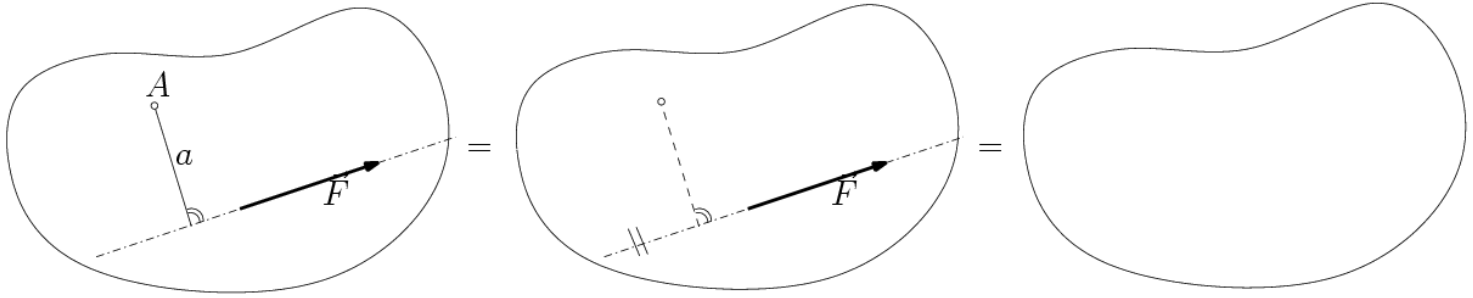
4.4.3. Moment as a vector

- The moment vector points in the direction in which a right-hand screw moves with the same direction of rotation



4.4.4. Parallel displacement of a force

- Moment action for displacement perpendicular to the line of action must be considered.



Force and moment

Box 4.14

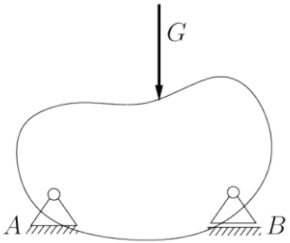
5. General Plane Structures

- Supporting structure is a system of elements whose mobility is limited. It is capable of absorbing forces.
- Structure elements (individual parts of the system):
  - Strut
  - Rope
  - Beam
  - Joint
  - Support
  - etc.
- „Static of rigid bodies“ → no deformations
- „Elasto static“→ Consideration of (small) deformations

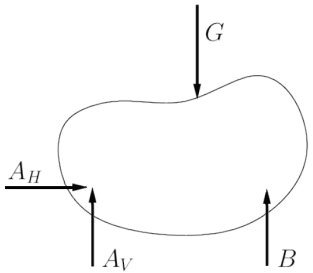
5.1. Cutting Principle

Destinction:

- **Active forces** are physically given, e.g., Weight force, wind, …
- **Reaction forces** result from impediments to freedom of movement, e.g., support forces
- Reaction forces must be **cut free** (make visible)!



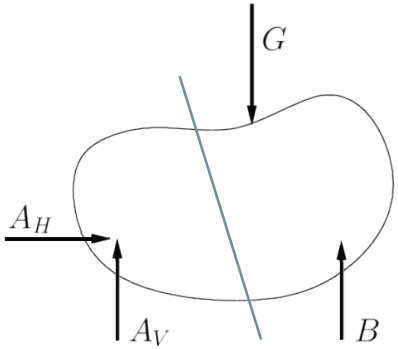
system



free−body diagram

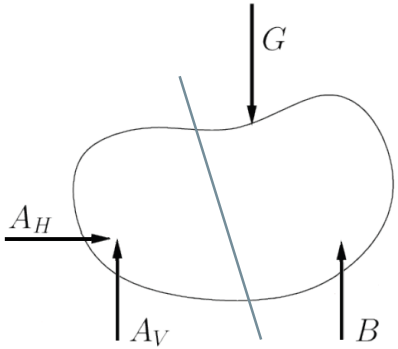
5.1. Cutting Principle

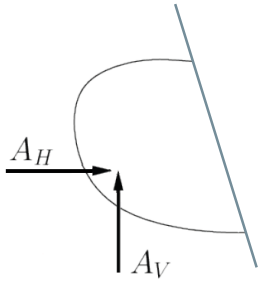
- The cutting principle is generally valid
- **Internal forces -** tensions - can also be exposed.

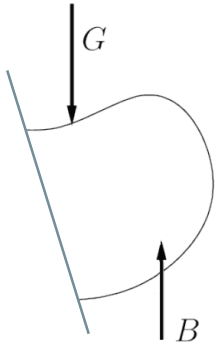


5.1. Cutting Principle

- The cutting principle is generally valid
- **Internal forces -** tensions - can also be exposed.

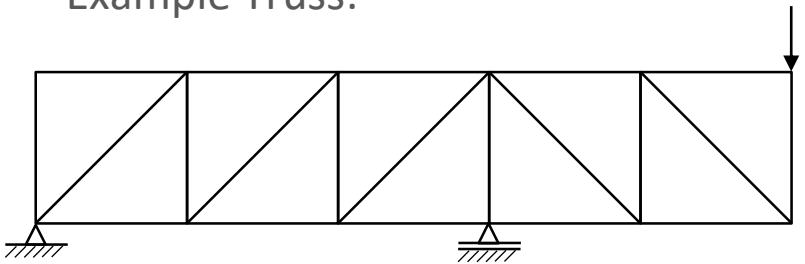






5.1. Cutting Principle

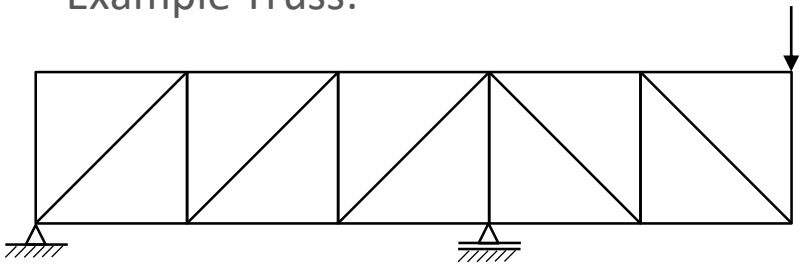
- Example Truss:





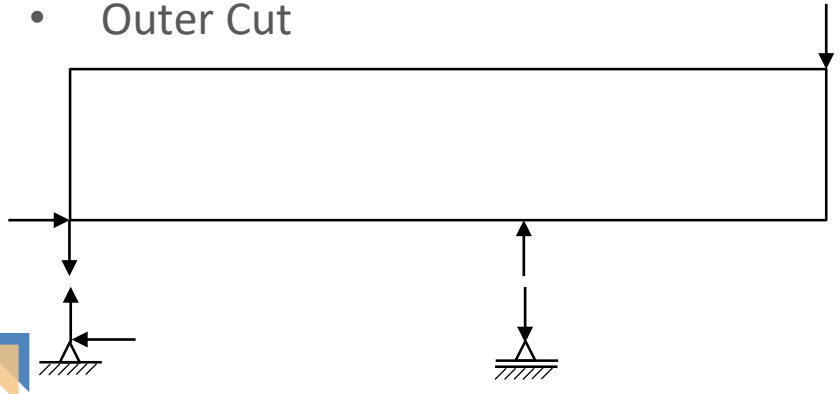
5.1. Cutting Principle

- Example Truss:



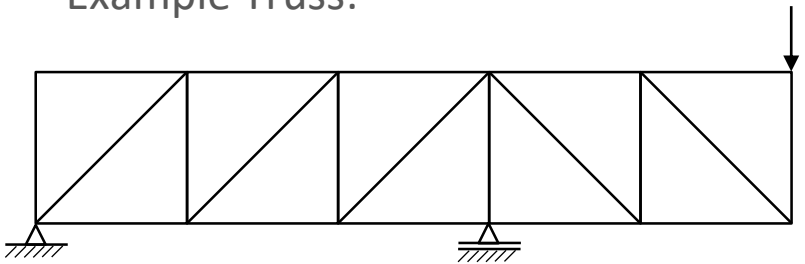


- Outer Cut



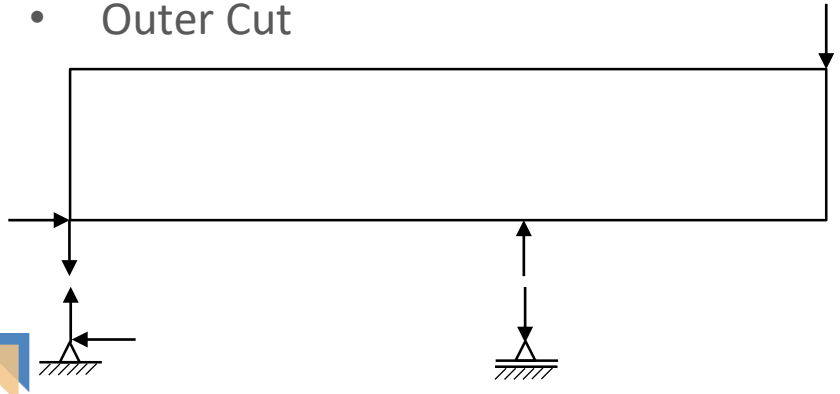
5.1. Cutting Principle

- Example Truss:

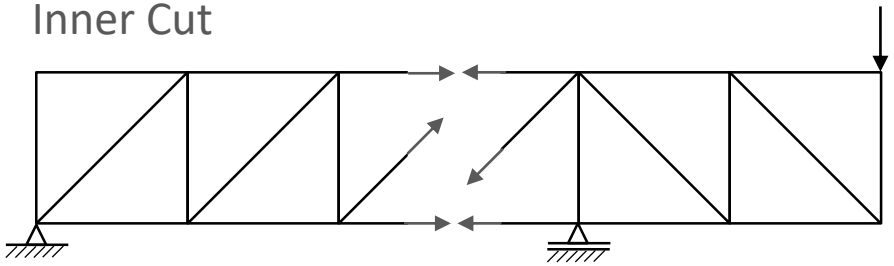




- Outer Cut

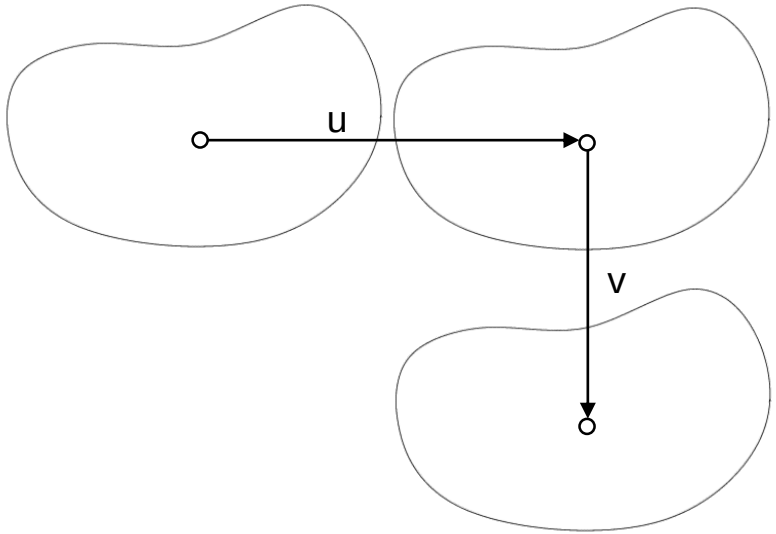


Inner Cut



5.2. Degrees of freedom and equilibrium conditions for plane structures

- 3 independent movement options in the plane:
  (degrees of freedom):

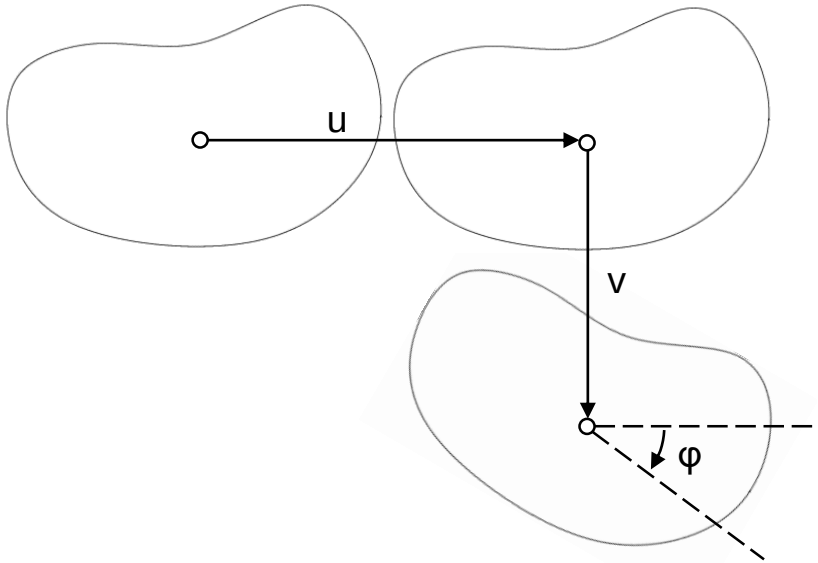


Box 5.1

5.2. Degrees of freedom and equilibrium conditions for plane structures

- 3 independent movement options in the plane (degrees of freedom):

Box 5.1



5.2. Degrees of freedom and equilibrium conditions for plane structures

- Motion in the plane must be constrained.

- 3 degrees of freedom lead to 3 equilibrium conditions:

Box 5.2

5.2.1. Alternative equilibrium conditions

- Attacking forces and moments can be changed via reduction, decomposition and parallel displacement:

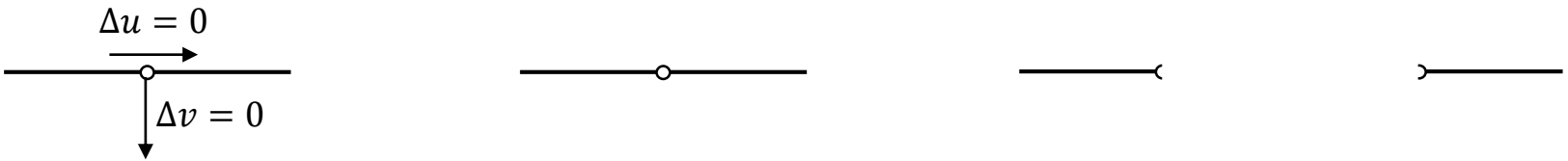
$$\sum M_A = 0$$ with arbitrary point A as center of rotation
$$\sum M_B = 0$$ with arbitrary point B ≠ A as center of rotation
$$\sum F_{\vec{e}_1} = 0$$ where $\overrightarrow{e_1}$ must not be perpendicular to the line AB

$$\sum M_A = 0$$ with arbitrary point A as center of rotation
$$\sum M_B = 0$$ with arbitrary point B ≠ A as center of rotation
$$\sum M_C = 0$$ where A, B & C must not lie on a line

5.3. Supports, support reactions, joints

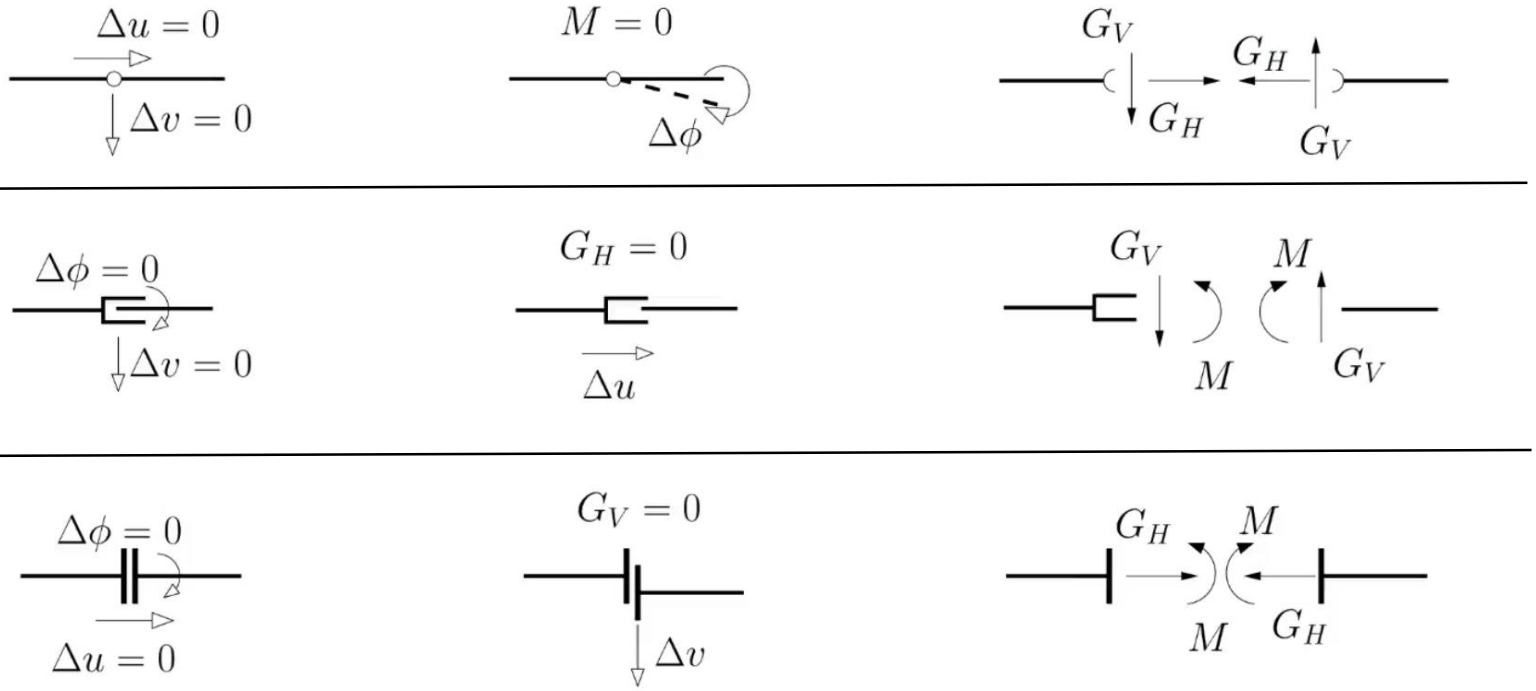
Joints

- **Connecting elements** of the support system
- Differentiation between possible and hindered degrees of freedom
- Example of a swivel joint (**moment joint**):



- Also **normal force** and **shear force** joint (next slide).

5.3. Supports, support reactions, joints



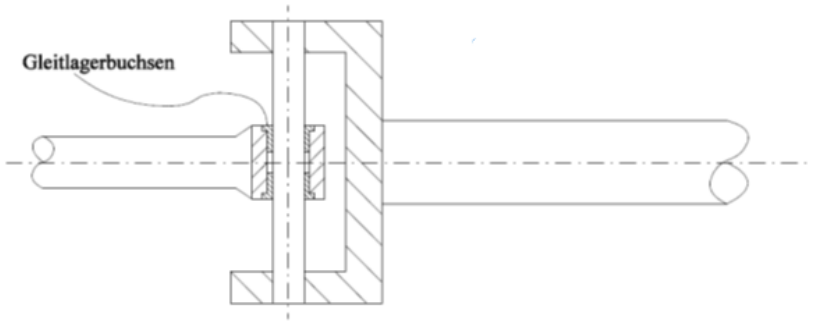
5.3. Supports, support reactions, joints



Real Joints

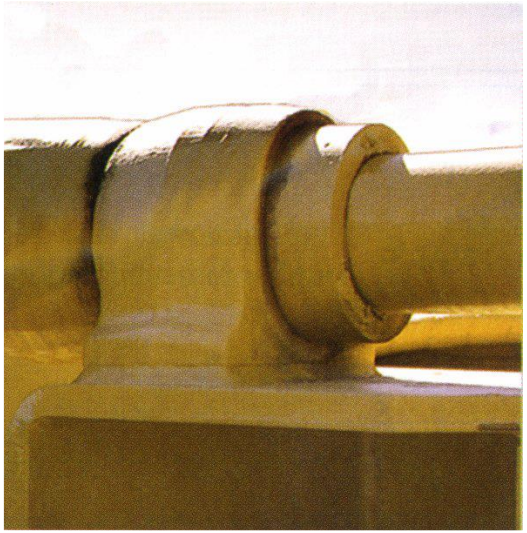


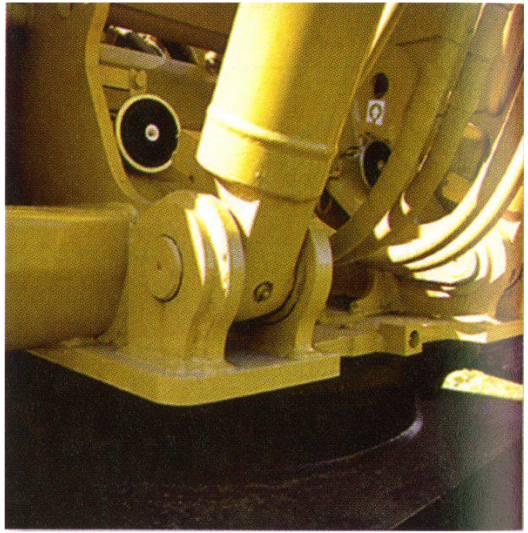




5.3. Supports, support reactions, joints

Real Joints

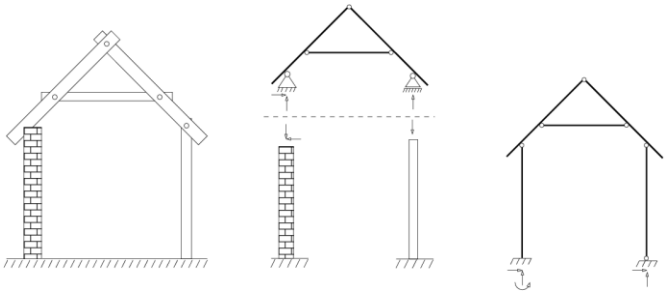






5.3. Supports, support reactions, joints

Supports and Support reactions



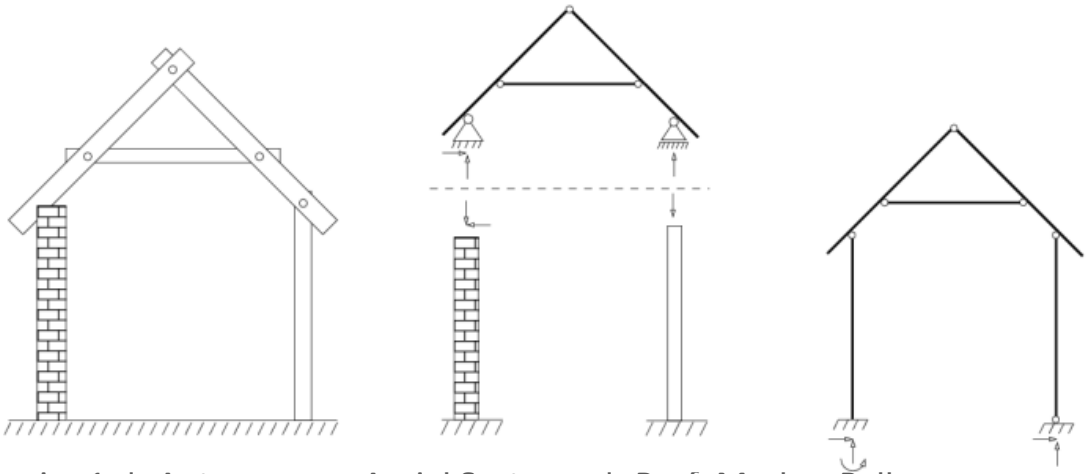
- connect supporting structures with their environment
- impede the movement of a system
- transfer forces to the environment
- delimit the system under consideration (e.g. walls as supports or the entire structure on its base).

Question of modeling!

5.3. Supports, support reactions, joints

Supports and Support reactions

- connect supporting structures with their environment
- impede the movement of a system
- transfer forces to the environment
- delimit the system under consideration (e.g. walls as supports or the entire structure on its base).

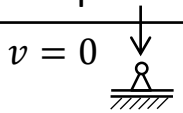
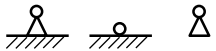
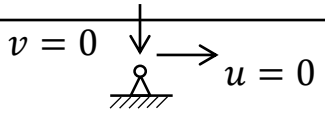
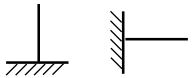
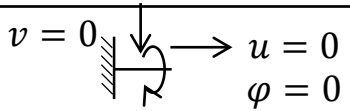
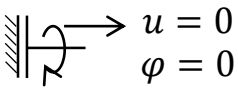
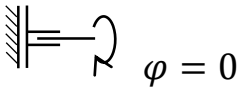
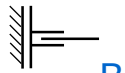


5.3. Supports, support reactions, joints

- Supports are differentiated according to their support type.
- Each constrained degree of freedom results in an associated force action.
- Force effects are made visible using the section principle.
- The value of a support is the number of impeded degrees of freedom (DoF).
- Symbols for each support type (following slide)

5.3. Supports, support reactions, joints

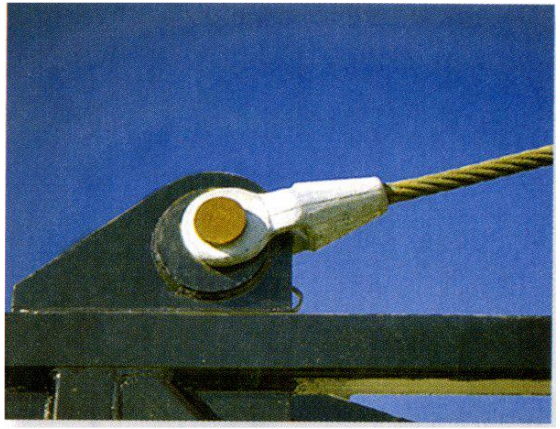
Table of plane support symbols

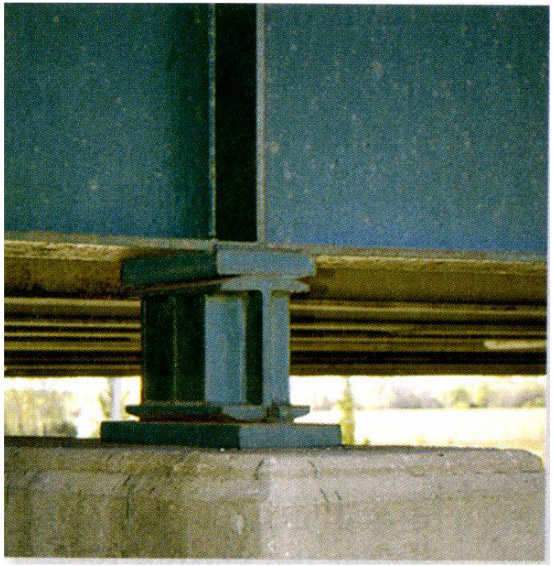
Name	Symbol	Value	Force action with impeded DoF	Unimpeded DoF
Sliding support		r = 1		
Fixed support		r = 2		
Clamping		r = 3		—
Displaceable clamping		r = 2		
Sliding sleeve		r = 1		

Box 5.3

5.3. Supports, support reactions, joints

Real Supports



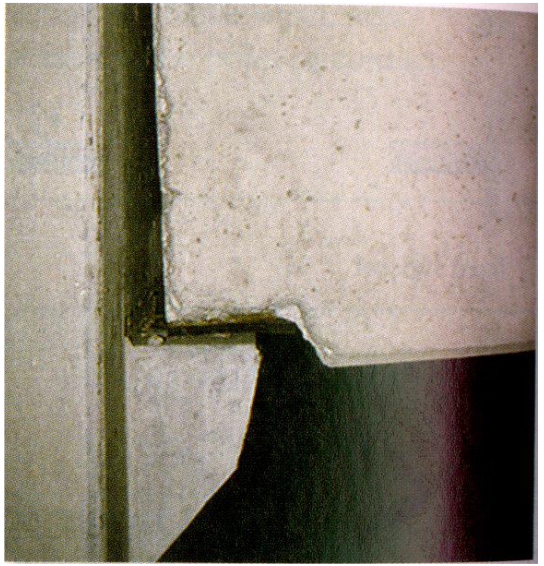




5.3. Supports, support reactions, joints

Real Supports





5.4. Static Determinacy

A System is **statically determined**, if

- all bearing reactions can be calculated from the equilibrium conditions.
- all degrees of freedom of the system are uniquely constrained.
- no pole exists for any structural element

➢ two necessary criteria: Counting criterion + Pole criterion

System is **statically indeterminate** if

- more support and intermediate conditions are present than equilibrium conditions

5.4.1. Counting criterion

Box 5.4

A = Number Support Reactions

Z = Number Intermediate Reactions

S = Number of elements (in 2D: **Disks**)

- Counting criterion is **necessary condition**
- Another necessary condition: **kinematic determinacy** (System not movable)

à Static determinacy only by **combination of both** conditions

5.4.1. Counting criterion

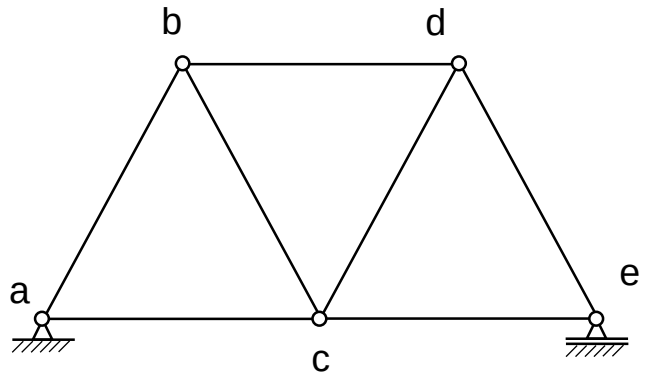
- Number of support reactions, corresponds to the value of the support.
- Value of a joint causes intermediate reactions
- For a moment/ball joint in 2 and 3 dimensions, respectively:



5.4.1. Counting criterion

Example 5.1: Counting criterion

Sketch:



Counting criterion:

Box 5.5

Quiz

- What is the minimum number of forces needed to impart a moment to a body without giving it a tendency to move?

- Find examples from engineering where force and moment are converted into each other.